

Guidelines

- Kindly use the given template for submitting your prototype(Make a copy of the template)
- One team is only required to submit one project.
- Follow this template while preparing your presentation.
- You are welcome to add as many POCs and design concepts to support your project.
- The project should be feasible and the team members should be capable enough, to come up with the fully functioning prototype of the same idea, if required.



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AI-Powered Product Intelligence for
Industrial Commerce

Team Details

Team Name: Tolerance

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Executive Brief: Tolerance

- Core Problem: Industrial distributor catalog data is highly cryptic (e.g. '3/8 CPLG BRS 150#'), misspelled, and lacks structure, rendering products unsearchable and driving up catalog maintenance costs.
- Our Solution: An AI-powered product data intelligence pipeline. Combines deterministic parsing rules for speed and accuracy with LLM-assisted classification, structured attribute extraction, and validation.
- Key Innovation: Exposes confidence scores and provenance tags (Rule-Based vs LLM-Inferred) for every single specification, providing a transparent audit trail for human catalogers.

Technical Challenge Answers

1. Understanding Limited Info (Enrichment):

- We utilize a hybrid model. Clean dimensions and materials are tokenized using fast deterministic regex patterns. Ambiguous text fragments are parsed using Gemini Flash by supplying the category context.

2. Accuracy & Trust (Validation):

- Tolerance runs sanity validation rules (e.g., flagging PVC material with class 150 metal ratings). Missing fields are filled using Gemini's general domain knowledge, labeled clearly as 'AI-Inferred' for easy auditing.

3. Scalability (Enterprise Catalog Engine):

- Built as a concurrent chunked batch pipeline. Repeat manufacturer name lookups are cached locally. In production, this scales via an asynchronous job queue (BullMQ/Redis) with multiple background workers.

Opportunities & USP

- Frictionless Audit Trail: Traditional AI classifiers behave as black boxes. Tolerance displays confidence margins and source provenance tags directly in the cataloger UI, enabling rapid auditing.
- Active Conflict Checking: Rather than outputting corrupt values, Tolerance flags logical contradictions (such as incompatible fitting sizes and thread types).
- Rate-Limit Immunity: Pre-cached seed catalog routes guarantee 100% demo availability and rate-limit immunity during live evaluation.

Features Offered by the Solution

- Fuzzy Manufacturer & Brand De-duplication: Reconciles raw spelling errors (e.g. 'lasco fittings ltd').
- 3-Level Deep Parts Taxonomy Classification: Places items into a structured category path using Gemini Flash.
- Structured Attribute Extractor: Extracts key-value-unit specifications.
- UOM & Fraction Standardization: Spacing, casing, and decimal-to-fraction conversions.
- E-commerce Description Builder: Generates CAPS invoice descriptions (≤ 40 chars) and mobile descriptions (60-80 chars).
- ROI Evaluation Module: Tracks data density gains and calculates manual analyst hours saved.

Process Flow Diagram

[Raw Messy Product Row] --> [Schema Profiler]

--> [Manufacturer & Brand De-duplication (Fuzzy Mapper)]

--> [Taxonomy Classifier (Gemini Flash)]

--> [Hybrid Attribute Extraction (Regex + LLM)]

--> [Validation & AI Enrichment (Conflict Check & Defaults)]

--> [UOM Normalization & Cleansing (Standard Spacing)]

--> [Invoice & Mobile Description Builder]

--> [Output: Enriched JSON/CSV + Business ROI Dashboard]

Wireframes/Mock diagrams of the proposed solution (optional)

Proposed Solution Architecture

- Client Dashboard: Next.js 15, TypeScript, Tailwind CSS v4, Lucide Icons.
- API Gateway: Next.js API Routes (/api/enrich) handling JSON payload processing.
- Core Engine: Structured TypeScript utility pipe coordinates all modules sequentially.
- LLM Provider: Google Gemini API via @google/generative-ai routing to gemini-flash-latest.
- Infrastructure: Vercel (Hobby Tier) for live public hosting.

Technologies Used

- Next.js 15 & React 19 (Serverless Web Framework)
- TypeScript (Type-safe business logic)
- Tailwind CSS v4 & PostCSS (Modern technical styling)
- Google Gemini Flash SDK (AI-assisted taxonomy & extraction)
- Vercel Platform (Continuous deployment & hosting)

Estimated Implementation Cost

- LLM Cost: ~\$0.000075 per SKU (using Gemini 1.5 Flash API pricing)
- Storage: \$0.00 (uses serverless local memory/state)
- Hosting: \$0.00 (Vercel Hobby Tier)
- Total Cost for MVP & Testing: \$0.00

Snapshots of the MVP

- Engineering Blueprint Transformation View: Floating schematic cards and leader lines.
- Specification Audit Trail: Interactive table mapping confidence and provenance.
- Business ROI Metrics Panel: Analytics cards showing manual hours saved.
- Batch comparative data table showing raw input vs resolved output.

Future Development & Roadmap

- ERP Connectors: Directly stream enriched data into popular distributor ERPs (e.g. Epicor P21, Infor, SAP).
- Digital Asset Sourcing: Automated scraping agents to locate and download PDF manual sheets and catalog drawings.
- Collaborative Human-in-the-loop (HITL): UI dashboard for catalogers to approve, reject, or correct AI extractions, retraining model rules in real-time.

Final Submission Links

GitHub Public Repository:

- <https://github.com/Saatvik-G/Tolerance.git>

Working Prototype URL:

- <https://tolerance-ten.vercel.app/>

Demo Video Link (3 Minutes):

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<https://app.guidde.com/share/playbooks/8iiUdixCsLkudTRbbf8Jws?origin=PBBuMYerseb1XleJip2pj6eLYmS2&mode=videoOnly>

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Thank You

